

DATA ENGINEERING

Nifty 50 Stocks

TABLE OF CONTENTS

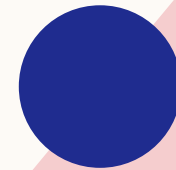
Introduction

How we get there?

Dataset

Visualization

Conclusion



INTRODUCTION

The Nifty 50 is a benchmark stock market index in India, representing the performance of the 50 largest and most liquid stocks listed on the National Stock Exchange (NSE). It is one of the key indicators of the Indian equity market and is widely used by investors, traders, and analysts to assess the overall health and trends of the stock market.

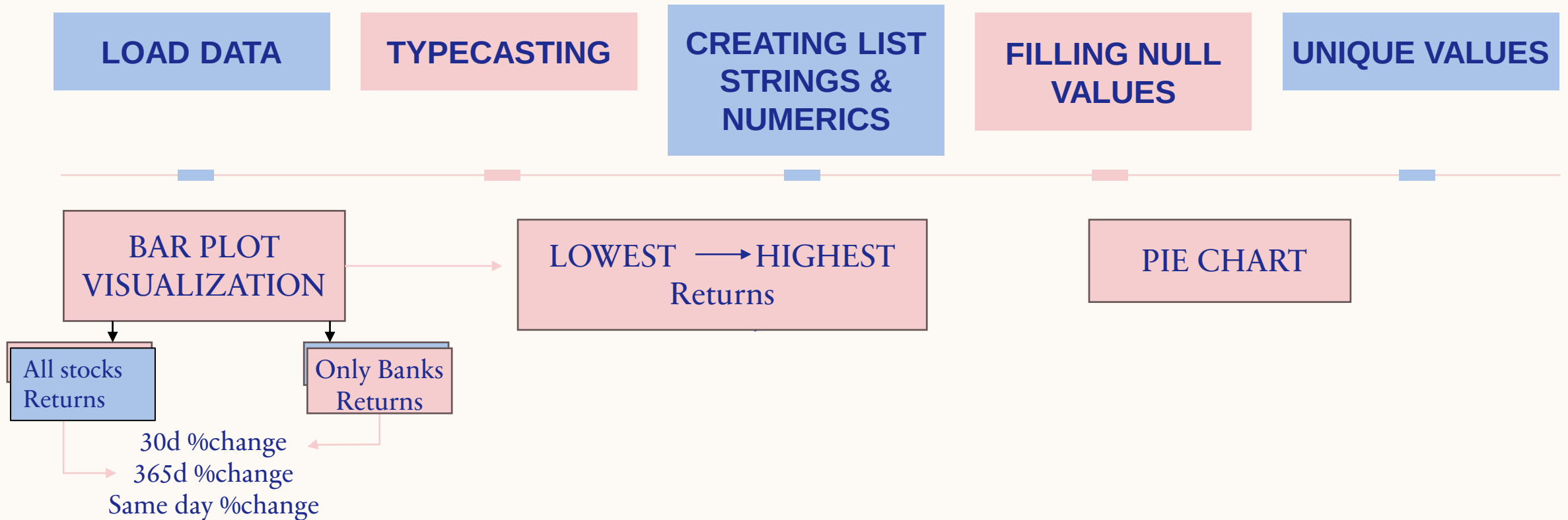
OBJECTIVE

Main Objective is to Understand and Analyze Nifty 50 Stocks

DATA SOURCE

NATIONAL STOCK EXCHANGE OF INDIA_LTD - NSE

HOW WE GET THERE



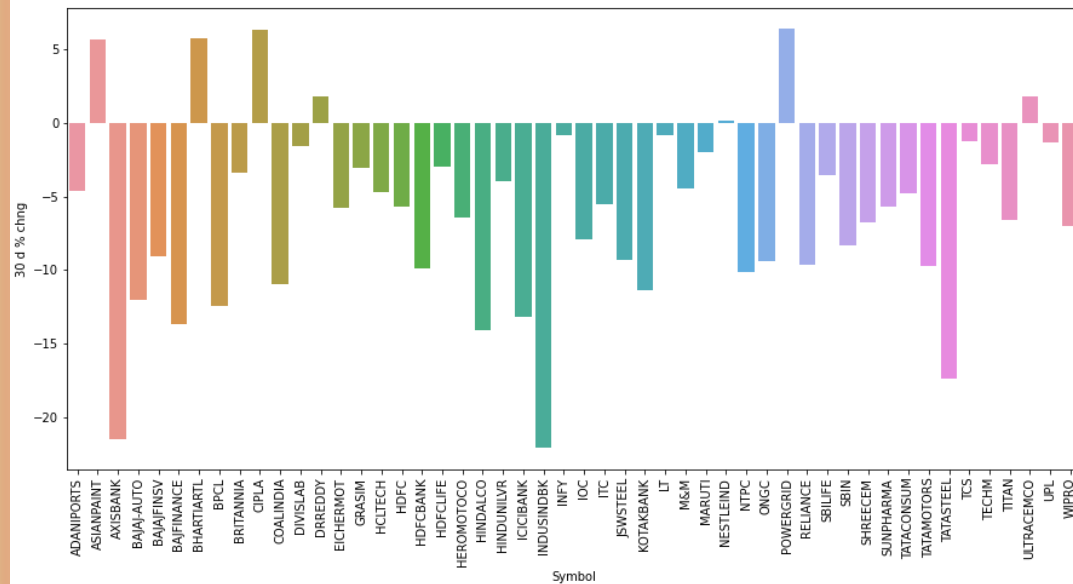
DATASET

Symbol	Open	High	Low	LTP	Chng	% Chng	Volume (lacs)	Turnover (crs.)	52w H	52w L	365 d % chng	30 d % chng
ADANIPTS	750	766	713.25	715	-47.45	-6.22	72.2	532.63	901	384.4	79.22	-4.65
ASIANPAINT	3,101.00	3,167.35	3,091.00	3,138.00	-6.25	-0.2	10.29	322.53	3,505.00	2,117.15	45.66	5.66
AXISBANK	669	674.9	660.45	661	-18.9	-2.78	102.53	684	866.9	568.4	10.19	-21.49
BAJAJ-AUTO	3,370.00	3,383.50	3,320.00	3,335.00	-56.7	-1.67	3.42	114.59	4,361.40	3,041.00	9.3	-12.05
BAJAJFINSV	17,200.00	17,237.20	16,610.00	16,684.00	-684.85	-3.94	3.42	576.79	19,325.00	8,273.70	91.38	-9.1
BAJFINANCE	7,021.00	7,047.90	6,775.00	6,780.00	-345.8	-4.85	16.89	1,161.63	8,050.00	4,362.00	44.57	-13.69
BHARTIARTL	763	763	733.1	735.85	-29.3	-3.83	111.43	830.06	781.8	454.11	58.55	5.7
BPCL	397.15	397.2	375	377.4	-22.7	-5.67	100.23	383.54	503	357	-1.22	-12.45
BRITANNIA	3,560.00	3,635.10	3,533.95	3,566.60	-6.8	-0.19	3.73	133.23	4,153.00	3,317.30	0.3	-3.42
CIPLA	892	976.05	890.65	965	65.05	7.23	144.59	1,380.90	1,005.00	726.5	31.89	6.34
COALINDIA	157.75	159.4	155.35	155.9	-2.65	-1.67	118.3	185.5	203.8	123.25	25.78	-10.94
DIVISLAB	4,770.00	5,077.70	4,756.75	4,940.00	140.2	2.92	15.71	775.37	5,425.10	3,153.30	42.39	-1.57
DRREDDY	4,580.00	4,820.00	4,576.15	4,750.00	158.4	3.45	10.72	508.97	5,614.60	4,135.00	-1.17	1.8
EICHERMOT	2,495.00	2,506.10	2,421.50	2,440.75	-79.65	-3.16	5.55	136.56	3,037.00	2,303.70	-5.95	-5.77
GRASIM	1,757.30	1,757.85	1,679.00	1,685.80	-80.95	-4.58	7.48	127.84	1,893.00	840.05	99.95	-3.08
HCLTECH	1,120.00	1,126.00	1,103.30	1,111.65	-13.15	-1.17	22.07	246.06	1,377.75	814.35	34.79	-4.73
HDFC	2,820.35	2,856.00	2,723.00	2,745.00	-122.75	-4.28	33.53	927.88	3,021.10	2,179.30	25.27	-5.72
HDFCBANK	1,500.00	1,506.70	1,485.00	1,489.50	-36.45	-2.39	93.12	1,394.10	1,725.00	1,342.00	6.18	-9.88
HDFCLIFE	685	689	667.1	669.75	-19.05	-2.77	22.37	151.4	775.65	617.4	0.7	-2.94
HEROMOTOCO	2,580.00	2,589.70	2,505.15	2,526.80	-67.9	-2.62	6.85	174.04	3,629.05	2,505.15	-16.02	-6.43
HINDALCO	441.8	442.7	414.7	417.7	-29.35	-6.57	148.26	631.93	551.85	220.35	86.93	-14.06
HINDUNILVR	2,344.00	2,365.00	2,325.20	2,340.90	-8.15	-0.35	24.51	572.85	2,839.30	2,120.00	9.6	-3.94
ICICIBANK	739	742.05	718.6	720.45	-30.6	-4.07	189.88	1,385.86	867	465.8	52.41	-13.14
INDUSINDBK	951	956.95	898	899.95	-59.35	-6.19	67.46	622.74	1,242.00	789	5.25	-22.08
INFY	1,702.55	1,718.35	1,684.00	1,689.55	-32.85	-1.91	44.94	764.67	1,848.00	1,091.00	51.44	-0.83
IOC	125.6	125.6	120.5	121.15	-4.5	-3.58	77.25	94.57	141.5	84	41.28	-7.87
ITC	228.9	230.05	223.1	223.6	-7.7	-3.33	270.27	610.54	265.3	192.4	15.35	-5.53
JSWSTEEL	668.25	672.55	624.25	630	-50.9	-7.48	89.22	574.61	776.5	336	86.25	-9.27
KOTAKBANK	2,002.00	2,007.00	1,955.10	1,960.00	-75.1	-3.69	26.48	522.52	2,253.00	1,626.00	5.24	-11.35
LT	1,820.00	1,841.75	1,768.60	1,781.00	-68.9	-3.72	27.97	502.81	1,981.75	1,092.00	59.59	-0.85
M&M	885	885	843	855.05	-36.15	-4.06	39.34	338.08	979	660.25	18.77	-4.42
MARUTI	7,520.00	7,520.00	7,130.00	7,150.00	-422.5	-5.58	11.55	840.81	8,368.00	6,400.00	1.34	-2.02
NESTLEIND	19,148.85	19,434.10	18,982.50	19,250.00	71.95	0.38	0.56	108.61	20,609.15	16,002.10	9.87	0.17
NTPC	133.2	134.05	128	128.65	-6.55	-4.84	133.24	173.94	152.1	88.15	36.93	-10.16
ONGC	152.25	152.25	146.25	147.75	-7.35	-4.74	231.36	344.33	172.75	77.05	82.86	-9.41
POWERGRID	204.05	204.95	200.8	202.5	-1.75	-0.86	96.11	195.09	209.95	136.88	3.69	6.36
RELIANCE	2,467.80	2,477.60	2,401.50	2,405.10	-87.85	-3.52	72.75	1,770.19	2,751.35	1,830.00	23.48	-9.62
SBI LIFE	1,154.00	1,154.00	1,105.25	1,130.85	-28.65	-2.47	23.16	262.43	1,273.90	825.2	33.19	-3.52
SBIN	486.25	487.9	467.1	470	-20.55	-4.19	263.06	1,249.55	542.3	240.15	93.42	-8.3
SHREECEM	26,450.00	26,539.90	25,812.00	25,900.00	-770.5	-2.89	0.3	76.94	32,048.00	22,531.00	9.29	-6.76
SUNPHARMA	775	798.9	762	767.25	-15.65	-2	54.33	424.05	851	502.3	51.57	-5.69
TATACONSUM	800.2	805	763.15	769.9	-37.9	-4.69	26.17	203.32	889	505.05	49.55	-4.82
TATAMOTORS	486	486.75	458	459.4	-33.35	-6.77	517.88	2,430.36	536.7	156.7	167.95	-9.68
TATASTEEL	1,157.90	1,159.50	1,106.25	1,110.25	-63.4	-5.4	106.46	1,200.79	1,534.50	539.5	105.13	-17.37
TCS	3,425.00	3,490.00	3,411.90	3,439.20	-6.7	-0.19	19.41	670.58	3,989.90	2,624.45	27.32	-1.25
TECHM	1,544.00	1,550.00	1,510.15	1,519.00	-40.35	-2.59	15.22	232.97	1,630.00	846.7	76.17	-2.83
TITAN	2,377.80	2,385.10	2,285.05	2,293.00	-104.8	-4.37	12.89	298.54	2,677.90	1,300.35	75.45	-6.59
ULTRACEMICO	7,550.00	7,599.00	7,370.10	7,398.45	-210.35	-2.76	2.66	198.32	8,269.00	4,770.00	53.5	1.78
UPL	726	726	701	703.5	-23.8	-3.27	24.82	176.35	864.7	414.15	68.06	-1.37
WIPRO	632	634.4	619.65	621.3	-15.4	-2.42	41.39	259.37	739.85	346.25	77.51	-7.01

```
plt.figure(figsize=(15,7))
plt.xticks(rotation=90)

sns.barplot(data = df, x = 'Symbol', y = '30 d % chng')
plt.show
```

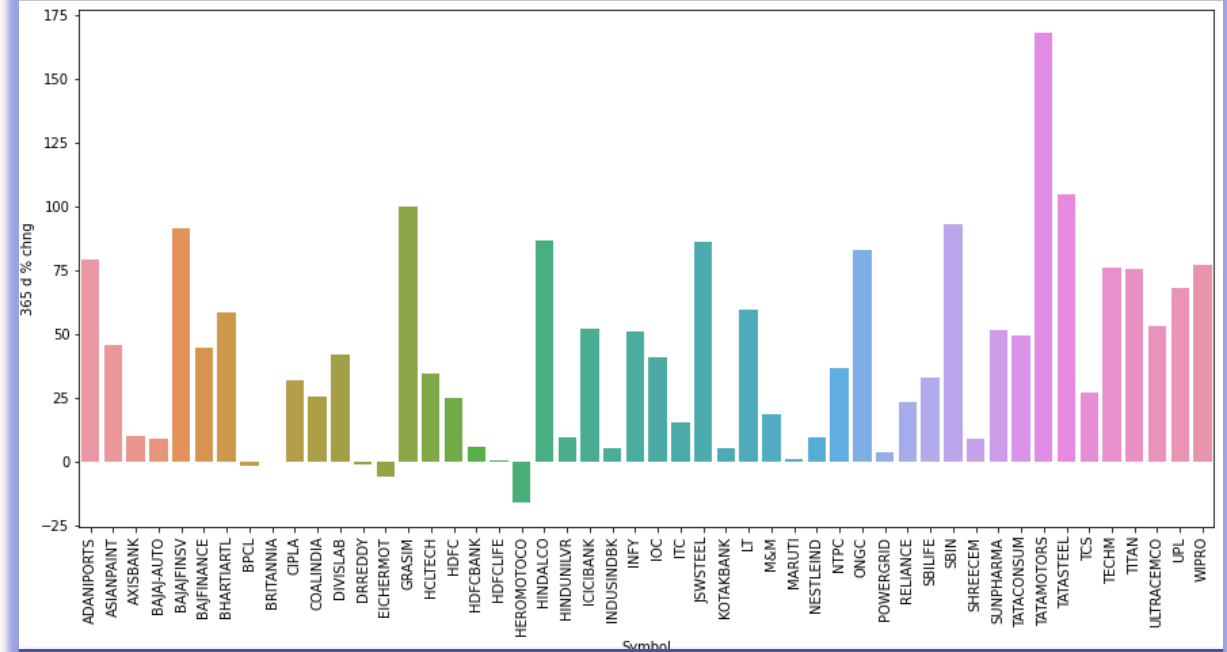
```
<function matplotlib.pyplot.show(close=None, block=None)>
```



By Plotting BarPlot, we can find that majority of the companies have shown negative returns if stayed invested for a month in it.

```
plt.figure(figsize=(15,7))
plt.xticks(rotation=90)
sns.barplot(data = df, x = 'Symbol', y = '365 d % chng')
plt.show
```

```
<function matplotlib.pyplot.show(close=None, block=None)>
```

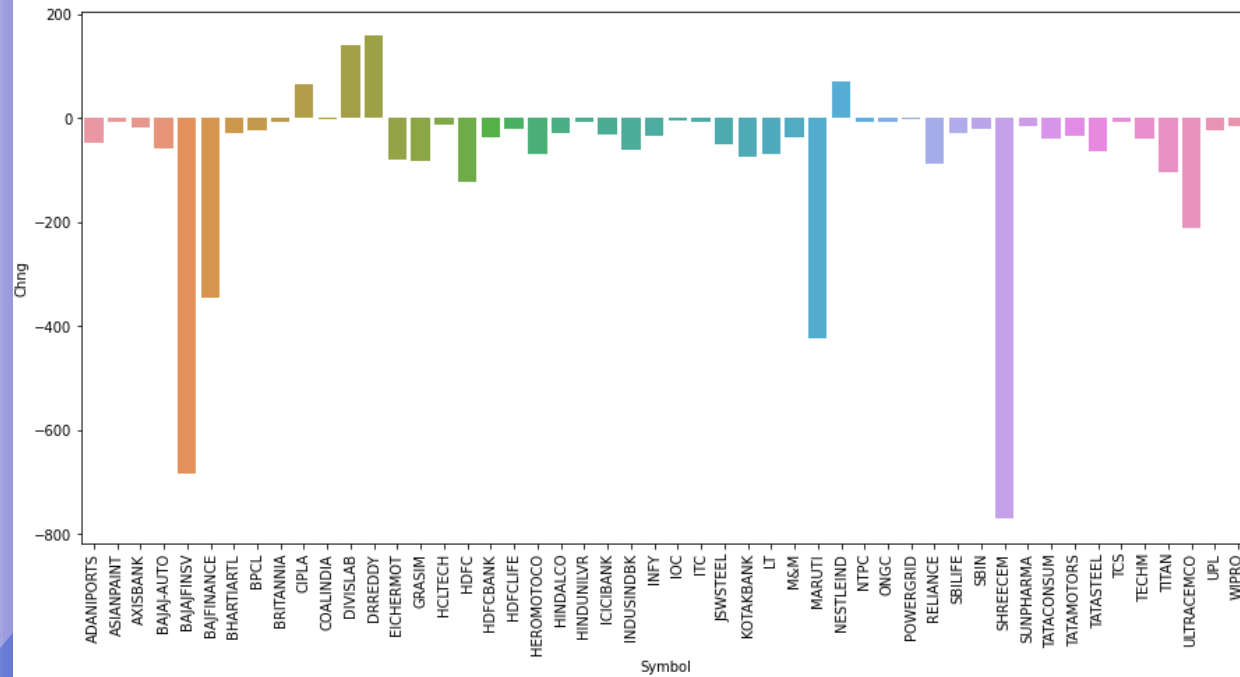


Plotting BarPlot, we can find that majority of the companies have shown atleast 10% returns if stayed invested for a year in it

```
plt.figure(figsize=(15,7))
plt.xticks(rotation=90)

sns.barplot(data = df, x = 'Symbol', y = 'Chng')
plt.show
```

```
<function matplotlib.pyplot.show(close=None, block=None)>
```

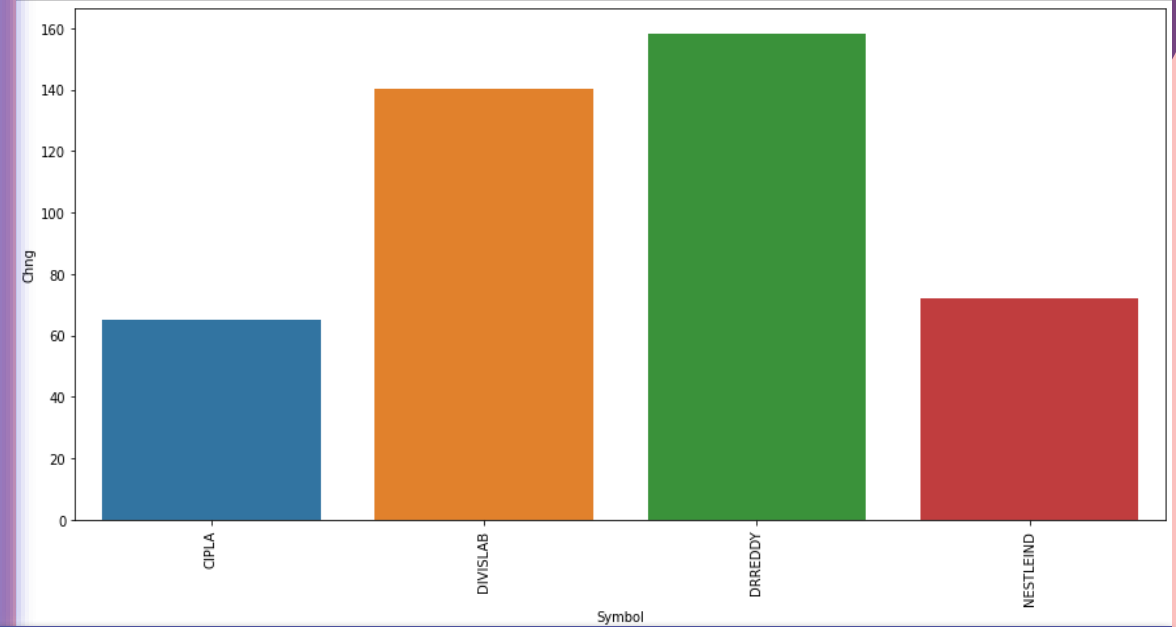


```
df[(df['Chng']>0)]
```

```
plt.figure(figsize=(15,7))
plt.xticks(rotation=90)
```

```
sns.barplot(data = df[(df['Chng']>0)], x = 'Symbol', y = 'Chng')
plt.show
```

```
<function matplotlib.pyplot.show(close=None, block=None)>
```



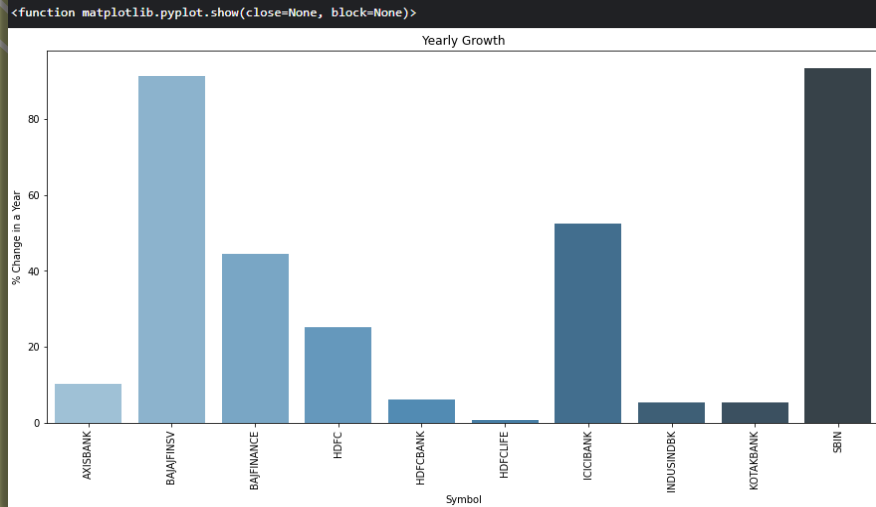
On the day for which the data is recorded, only 4 companies gave positive returns

```
plt.figure(figsize=(15,7))
plt.xticks(rotation=90)

sns.barplot(data = bank_table, x = 'Symbol', y = '365 d % chng', palette='Blues_d')

plt.xlabel("Symbol")
plt.ylabel('% Change in a Year')
plt.title("Yearly Growth")

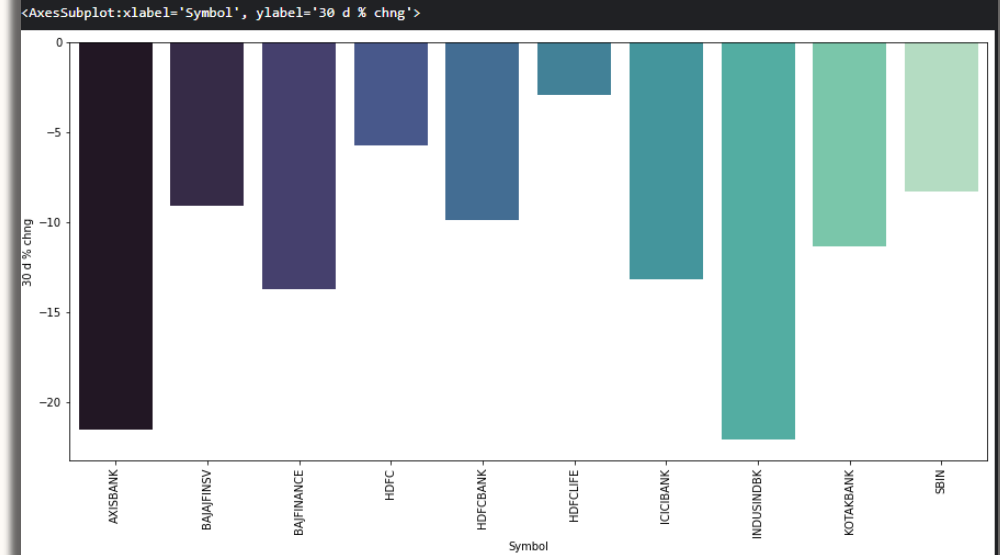
plt.show
```



BajajFinsv, BajajFinance,IciciBank,SBIN outperformed in Banking sector, giving more than 40% returns in a Year

```
plt.figure(figsize=(15,7))
plt.xticks(rotation=90)

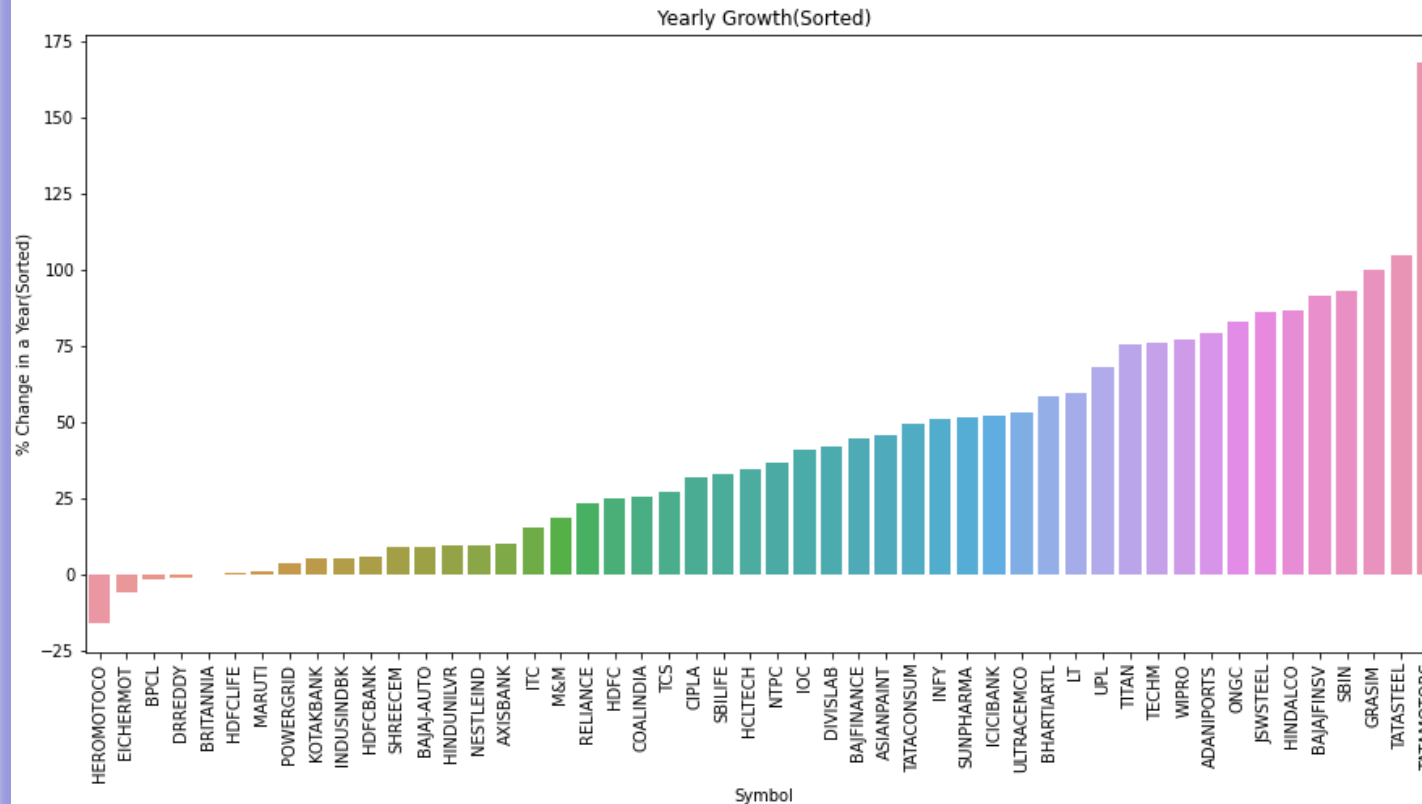
sns.barplot(data = bank_table, x = 'Symbol', y = '30 d % chng',palette='mako')
```



AxisBank and IndusInd Bank are the worst performers with more than 20% loss of the investors in a single month.


```
plt.figure(figsize=(15,7))
plt.xticks(rotation=90)
sns.barplot(data = sort_yearly, x = 'Symbol', y = '365 d % chng')
plt.xlabel("Symbol")
plt.ylabel('% Change in a Year(Sorted)')
plt.title("Yearly Growth(Sorted)")
plt.show
```

```
<function matplotlib.pyplot.show(close=None, block=None)>
```



TataMotors gave the highest returns in a year, more than 167.95%
HeroMotoCo gave the lowest returns in a year, less than -16.02%

SUMMARY

1. Investment Duration Insights:

- By analyzing Bar Plots, it is evident that a majority of the companies have shown negative returns if held for a month. Therefore, it is recommended to consider an investment period longer than a month.
- For a one-year investment horizon, a significant number of companies displayed positive returns, with some even exceeding 10%. This suggests that longer-term investments may yield more favorable results.

2. Daily Returns Analysis:

- On the recorded day, only a small fraction of the companies exhibited positive returns, indicating a mixed performance in the market on that specific day.

3. Sector-Specific Performance

- Notable outperformance was observed in the Banking sector, with companies like BajajFinsv, BajajFinance, ICICI Bank, and SBIN providing returns exceeding 40% in a year.
- Conversely, AxisBank and IndusInd Bank were identified as the worst performers, resulting in losses exceeding 20% for investors in a single month.

4. Individual Stock Performance

- TataMotors emerged as the top performer, delivering the highest returns in a year, surpassing 167.95%.
- HeroMotoCo exhibited the lowest returns in a year, with a negative return of less than -16.02%.

5. Stock Volume Insights:

- By visualizing a pie chart, it is evident that Axis Bank had the highest volume of stocks sold, highlighting its significance and activity in the market.

